

Per-Image Decorrelation Efficiency of the BT.601 Fixed Transform

Measured Against KLT Ceiling Across the Kodak Lossless True Color Image Suite

Residual Correlation Structure and Chroma Subsampling Cost in PCD0992

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Abstract

The per-image PCA baselines established in Baetzel (2026a) documented the inter-channel redundancy structure of all 24 images in the Kodak Lossless True Color Image Suite, establishing the KLT ceiling at residual correlation $r = 0.000000$. This paper measures the decorrelation actually achieved by BT.601 YCbCr — the colorspace conversion deployed in virtually every lossy compression pipeline in active use — against that ceiling on each image individually. The central finding is structural: BT.601 leaves a mean residual correlation of 0.372 (range 0.150 to 0.669), achieving between 31.5% and 82.3% of the KLT optimum, but the dominant source of that residual is the Cb–Cr chroma pair, which retains a mean absolute correlation of 0.616 (range 0.299 to 0.934). This chrominance–chrominance redundancy is not addressed by the luminance–chrominance decomposition and passes largely untouched through standard channel-independent encoder stages — a measurable, persistent inefficiency in every JPEG-family pipeline since 1992. Per-image 4:2:0 chroma subsampling simulation produces PSNR values spanning 8.00 dB (39.91 to 47.91 dB), with information loss correlating to the blue channel independence metrics established in Baetzel (2026a). These measurements constitute the first published per-image quantification of the decorrelation gap between a deployed fixed transform and the KLT optimum across the full Kodak benchmark.

1 Introduction

The BT.601 coefficient matrix has served as the decorrelation step in JPEG and its successors since 1992 [1], converting RGB pixel data into YCbCr using fixed coefficients derived from broadcast television luminance characteristics [2] rather than from any property of the image being compressed. The KLT computes an image-specific rotation from each image’s covariance structure and achieves residual inter-channel correlation of exactly zero, but its per-image overhead has prevented practical deployment.

The gap between deployed fixed-coefficient decorrelation and the KLT optimum has been understood in principle but never quantified per-image across the Kodak Lossless True Color Image Suite, despite the suite’s status as the primary benchmark in compression research. Published evaluations have relied on suite-averaged statistics that collapse substantial inter-image variation into single aggregate measures.

Baetzel (2026a) addressed the characterization side by computing the first per-image PCA decomposition of all 24 images, revealing condition numbers spanning 7.55 to 1739.16 and blue

channel independence from 2.3% to 52.0% — establishing the per-image decorrelation limit against which the present paper now measures what BT.601 actually achieves.

The results reveal that the dominant residual is not between luminance and chrominance — where the transform concentrates its effort — but between the two chrominance channels themselves, a redundancy the fixed transform was never designed to address. BT.601 removes a mean of 55.4% of available correlation, but the remainder constitutes a measurable, systematic inefficiency that has persisted without per-image quantification for over three decades.

This paper analyzes each of the 24 Kodak images to quantify how much inter-channel decorrelation BT.601 captures, the nature of the residual structure, and the specific costs incurred during 4:2:0 chroma subsampling.

2 Methodology

2.1 Image Source

All 24 images from the Kodak Lossless True Color Image Suite were analyzed in their standard 768×512 base-resolution PNG format, identical to the source data used in Baetzel (2026a). The imaging chain provenance documented in that work — film spectral sensitivity, PCD Film Scanner 2000 trilinear CCD capture, PhotoYCC encoding and decode to 8-bit RGB — applies equally here.

2.2 BT.601 YCbCr Transform

The RGB-to-YCbCr conversion was performed using the ITU-R BT.601 coefficient matrix as specified in the JPEG standard [1, 2]:

$$\begin{aligned} Y &= 0.299R + 0.587G + 0.114B \\ C_b &= -0.168736R - 0.331264G + 0.5B + 128 \\ C_r &= 0.5R - 0.418688G - 0.081312B + 128 \end{aligned}$$

These coefficients are fixed across all images, derived from the luminance characteristics of standard broadcast television signals rather than from any property of the images being transformed. The transform was applied to the raw 8-bit pixel values in floating-point precision with no rounding or clipping prior to correlation measurement.

2.3 Computed Metrics

For each of the 24 images, the following measurements were computed in a single pass from the raw RGB pixel values, treating each pixel as a three-component observation vector:

PCA baseline replication: The complete eigendecomposition from Baetzel (2026a) — eigenvalues, eigenvectors, variance explained, condition number, blue channel independence, and dimensionality tier — was recomputed to ensure exact correspondence between baseline and gap measurements within each JSON output.

YCbCr residual correlations: Pairwise Pearson correlation coefficients between all three channel pairs in YCbCr space (Y–Cb, Y–Cr, Cb–Cr), computed after applying BT.601 to the full pixel array.

Decorrelation gap: The average absolute residual correlation after BT.601. Because the KLT achieves zero residual correlation by construction, the post-BT.601 correlation directly defines the gap.

Decorrelation efficiency: The fraction of original RGB correlation removed, computed as $(\text{RGB avg } |r| - \text{YCbCr avg } |r|) / \text{RGB avg } |r| \times 100$.

4:2:0 chroma subsampling PSNR: Peak signal-to-noise ratio between the original RGB image and a reconstruction obtained by converting to YCbCr, downsampling Cb and Cr by 2× in both dimensions using box averaging, upsampling to full resolution using nearest-neighbor interpolation, and applying the BT.601 inverse. Per-channel PSNR (R, G, B) was also computed.

2.4 Verification and Reproducibility

All PCA baseline values recomputed in this analysis match those reported in Baetzel (2026a) to the published precision, confirming identical source data and methodology across both papers. Per-image results are provided as machine-readable JSON files containing the complete measurement set. The analysis requires only Python 3 with NumPy and Pillow, operating on publicly available Kodak PNG files.

3 Results

3.1 Decorrelation Gap Across the Full Suite

Table 1 presents the complete decorrelation gap measurement for all 24 images in the suite. Each row reports the baseline average absolute correlation in RGB space, the residual average absolute correlation after BT.601 YCbCr conversion, the decorrelation efficiency as a percentage of the theoretical optimum, and the overall 4:2:0 chroma subsampling PSNR.

Table 1. Per-Image Decorrelation Gap: BT.601 vs. KLT Ceiling

Image	RGB Avg $ r $	YCbCr Avg $ r $	Gap %	Efficiency	Cb–Cr	PSNR 4:2:0	Tier
kodim01	0.8980	0.4221	47.0%	53.0%	-0.6600	45.07 dB	S 1-D
kodim02	0.6361	0.3989	62.7%	37.3%	-0.9338	42.31 dB	3-D
kodim03	0.5203	0.2956	56.8%	43.2%	-0.4060	44.02 dB	3-D
kodim04	0.7474	0.4997	66.9%	33.1%	-0.5502	43.8 dB	W 1-D
kodim05	0.8743	0.2689	30.8%	69.2%	-0.2993	41.36 dB	W 1-D
kodim06	0.9762	0.6691	68.5%	31.5%	-0.7158	45.21 dB	N-D
kodim07	0.8349	0.2039	24.4%	75.6%	-0.4947	44.29 dB	W 1-D
kodim08	0.9524	0.3888	40.8%	59.2%	-0.6290	42.75 dB	S 1-D
kodim09	0.8458	0.4353	51.5%	48.5%	-0.6954	45.52 dB	W 1-D
kodim10	0.9384	0.4071	43.4%	56.6%	-0.6249	45.67 dB	S 1-D

kodim11	0.8625	0.3312	38.4%	61.6%	-0.6603	45.03 dB	W 1-D
kodim12	0.9176	0.3497	38.1%	61.9%	-0.4396	46.18 dB	S 1-D
kodim13	0.9642	0.2091	21.7%	78.3%	-0.4237	43.04 dB	N-D
kodim14	0.6763	0.4297	63.5%	36.5%	-0.6295	39.91 dB	2-D
kodim15	0.9015	0.3810	42.3%	57.7%	-0.5451	43.25 dB	S 1-D
kodim16	0.9425	0.4280	45.4%	54.6%	-0.8545	47.91 dB	S 1-D
kodim17	0.9654	0.4326	44.8%	55.2%	-0.8160	46.11 dB	N-D
kodim18	0.8345	0.5663	67.9%	32.1%	-0.6084	42.48 dB	W 1-D
kodim19	0.8991	0.4830	53.7%	46.3%	-0.8690	45.95 dB	S 1-D
kodim20	0.9768	0.3566	36.5%	63.5%	-0.8708	43.92 dB	N-D
kodim21	0.8254	0.3860	46.8%	53.2%	-0.9192	44.83 dB	W 1-D
kodim22	0.8489	0.1498	17.7%	82.3%	-0.3327	42.57 dB	W 1-D
kodim23	0.5595	0.1966	35.1%	64.9%	-0.3459	42.37 dB	3-D
kodim24	0.9600	0.2377	24.8%	75.2%	-0.4508	40.1 dB	N-D
MEAN	0.8483	0.3720	44.6%	55.4%	-0.6156	43.90 dB	—

Across the full suite, BT.601 reduces average inter-channel correlation from 0.848 to 0.372, achieving a mean decorrelation efficiency of 55.4% — ranging from 31.5% (kodim06) to 82.3% (kodim22). This nearly threefold variation across images sharing the same fixed transform represents a systematic performance spread that suite-averaged reporting has historically concealed.

3.2 Cb–Cr Residual Correlation Structure

The dominant contributor to the decorrelation gap is the Cb–Cr channel pair. Table 2 presents the full residual correlation profile for all 24 images, separating the three YCbCr channel pairs to reveal the structure of what BT.601 leaves behind.

Table 2. YCbCr Residual Correlation by Channel Pair

Image	Y–Cb	Y–Cr	Cb–Cr	Avg r Residual
kodim01	-0.0856	-0.5207	-0.6600	0.4221
kodim02	-0.2038	0.0591	-0.9338	0.3989
kodim03	-0.3883	-0.0925	-0.4060	0.2956
kodim04	0.4880	-0.4609	-0.5502	0.4997
kodim05	-0.3854	-0.1219	-0.2993	0.2689
kodim06	-0.8502	0.4413	-0.7158	0.6691
kodim07	-0.0720	0.0449	-0.4947	0.2039
kodim08	-0.4779	-0.0596	-0.6290	0.3888
kodim09	-0.4896	0.1209	-0.6954	0.4353
kodim10	-0.5907	-0.0056	-0.6249	0.4071
kodim11	-0.0166	-0.3167	-0.6603	0.3312

kodim12	-0.1005	-0.5092	-0.4396	0.3497
kodim13	0.0207	-0.1830	-0.4237	0.2091
kodim14	-0.5458	0.1138	-0.6295	0.4297
kodim15	-0.0802	-0.5175	-0.5451	0.3810
kodim16	-0.3919	0.0376	-0.8545	0.4280
kodim17	-0.1756	-0.3062	-0.8160	0.4326
kodim18	-0.6075	0.4832	-0.6084	0.5663
kodim19	-0.2792	0.3006	-0.8690	0.4830
kodim20	0.0873	-0.1116	-0.8708	0.3566
kodim21	0.0018	-0.2371	-0.9192	0.3860
kodim22	0.0573	0.0595	-0.3327	0.1498
kodim23	-0.2343	0.0095	-0.3459	0.1966
kodim24	-0.1577	-0.1048	-0.4508	0.2377
MEAN r 	0.2828	0.2174	0.6156	0.3720

The luminance–chrominance pairs show moderate residual correlations (Y–Cb mean $|r| = 0.283$, Y–Cr mean $|r| = 0.217$), indicating partial but incomplete separation. The Cb–Cr pair dominates the residual structure, retaining a mean absolute correlation of 0.616, with individual images reaching 0.934 (kodim02) and 0.919 (kodim21). In 23 of 24 images, Cb–Cr exceeds both luminance–chrominance residuals, often substantially.

This pattern is structural. BT.601 was designed to separate luminance from chrominance — to extract the brightness signal and relegate color-difference signals to channels amenable to aggressive compression. It was not designed to decorrelate the chrominance channels from each other. The Cb–Cr residual is therefore not a failure relative to the transform’s original design intent, but it constitutes a substantial source of inter-channel redundancy that persists through the entire compression pipeline — redundancy that channel-independent quantization and entropy coding stages cannot exploit.

3.3 Decorrelation Gap by Dimensionality Tier

The dimensionality tier classification established in Baetzel (2026a) divided the 24 images into five groups based on the fraction of variance captured by their first principal component. Table 3 reports the mean decorrelation gap and mean 4:2:0 PSNR within each tier.

Table 3. Decorrelation Gap and Subsampling Cost by Dimensionality Tier

Tier	n	Mean RGB $ r $	Mean YCbCr $ r $	Mean Gap %	Mean PSNR
Three-dimensional	3	0.5720	0.2970	51.6%	42.90 dB
Two-dimensional	1	0.6763	0.4297	63.5%	39.91 dB
Weakly one-dimensional	8	0.8342	0.3551	43.0%	43.73 dB
Strongly one-dimensional	7	0.9214	0.4085	44.4%	45.25 dB
Near-degenerate	5	0.9685	0.3810	39.3%	43.68 dB

The relationship between dimensionality and decorrelation efficiency is not monotonic. The two-dimensional tier (kodim14 alone) produces the highest gap percentage at 63.5%, while the near-degenerate tier averages the lowest at 39.3%. This reflects a tension between two factors: images with very high baseline correlations concentrate variance along a single axis that the Y channel approximates well, so the fixed transform captures a larger fraction of available redundancy despite the greater absolute amount present. Images with moderate correlations present more complex multi-axis structures where the fixed coefficients may or may not align with the actual variance distribution, producing a wider spread of outcomes.

3.4 Per-Channel 4:2:0 Chroma Subsampling Cost

Table 4 reports the per-channel PSNR from 4:2:0 chroma subsampling simulation for all 24 images, revealing which channel absorbs the greatest information loss under the standard subsampling operation.

Table 4. Per-Channel 4:2:0 Chroma Subsampling PSNR (dB)

Image	R (dB)	G (dB)	B (dB)	Overall (dB)	Blue Indep.
kodim01	43.69 dB	48.47 dB	44.38 dB	45.07 dB	2.7%
kodim02	39.87 dB	46.86 dB	42.79 dB	42.31 dB	27.4%
kodim03	44.28 dB	48.16 dB	41.78 dB	44.02 dB	52%
kodim04	40.97 dB	46.78 dB	46.17 dB	43.8 dB	2.7%
kodim05	41.56 dB	46.35 dB	38.99 dB	41.36 dB	11.4%
kodim06	46.26 dB	50.15 dB	42.44 dB	45.21 dB	2.4%
kodim07	44.48 dB	48.73 dB	42.03 dB	44.29 dB	11.4%
kodim08	42.34 dB	47.62 dB	40.77 dB	42.75 dB	4.7%
kodim09	46.32 dB	49.79 dB	42.99 dB	45.52 dB	22%
kodim10	45.85 dB	49.56 dB	43.55 dB	45.67 dB	6.4%
kodim11	43.46 dB	48.82 dB	44.43 dB	45.03 dB	5%

kodim12	45.42 dB	49.26 dB	44.97 dB	46.18 dB	4.2%
kodim13	45.81 dB	49.11 dB	39.57 dB	43.04 dB	3.4%
kodim14	39.19 dB	45.87 dB	37.95 dB	39.91 dB	47.6%
kodim15	40.99 dB	46.51 dB	43.97 dB	43.25 dB	2.3%
kodim16	48.64 dB	51.02 dB	45.68 dB	47.91 dB	7.5%
kodim17	47.07 dB	50.52 dB	43.48 dB	46.11 dB	3.6%
kodim18	43.18 dB	47.87 dB	39.79 dB	42.48 dB	25.1%
kodim19	46.14 dB	49.91 dB	43.8 dB	45.95 dB	13%
kodim20	46.09 dB	49.91 dB	40.61 dB	43.92 dB	2.6%
kodim21	45.92 dB	49.95 dB	42.01 dB	44.83 dB	13%
kodim22	42.55 dB	46.73 dB	40.5 dB	42.57 dB	10.5%
kodim23	42.4 dB	48.1 dB	39.97 dB	42.37 dB	35.9%
kodim24	42.14 dB	46.88 dB	36.75 dB	40.1 dB	3.8%

The green channel consistently sustains the least subsampling damage (mean PSNR 48.42 dB), the blue channel the most (41.93 dB), with red between them (43.79 dB) — an ordering that mirrors the BT.601 luminance weights exactly (G: 0.587, R: 0.299, B: 0.114). The channel receiving the most Y-channel protection loses the least from chroma subsampling; the channel receiving the least protection absorbs the most damage.

The overall PSNR spans 8.00 dB across the suite, from 39.91 dB (kodim14) to 47.91 dB (kodim16), meaning the same subsampling operation costs nearly four times as much in mean squared error on the worst-case image compared to the best. This variation tracks the blue channel independence metric from Baetzel (2026a): kodim14, with a blue independence of 47.6%, carries nearly half its blue channel variance orthogonal to the luminance axis, leaving it unprotected during chroma subsampling. Kodim16, at 7.5% blue independence, loses comparatively little.

4 Discussion

These measurements quantify a systematic inefficiency that has been present in every JPEG-family compression pipeline since 1992. BT.601 achieves a mean decorrelation efficiency of 55.4%, meaning that on average, nearly half of the available inter-channel redundancy passes largely untouched through the colorspace conversion and enters the quantization and entropy coding stages still correlated.

The practical consequence is architectural. In standard JPEG encoding, the DCT operates independently on each channel after YCbCr conversion. Residual inter-channel correlation — particularly the Cb–Cr redundancy documented here — represents information that standard channel-independent encoding stages largely cannot reach. The identical quantization tables typically applied to Cb and Cr assume statistical independence between the two chrominance channels — a direct mismatch with the measured Cb–Cr correlations of 0.299 to 0.934, which

indicate that these channels retain substantial shared structure that uniform quantization does not account for.

The per-image nature of these measurements is significant given the finding in Baetzel (2026a) that the Kodak suite was curated to span the full range of inter-channel redundancy configurations. The five-tier dimensionality classification provides a framework for understanding the performance variation: images whose color-space geometry happens to align with the BT.601 coefficient structure yield high decorrelation efficiency, while images whose principal variance axes diverge from the fixed luminance direction yield low efficiency and correspondingly higher residual correlation. Because published benchmarks have reported suite-averaged metrics, this variation has been collapsed into aggregate statistics that obscure the per-image performance profile of the decorrelation step.

The 4:2:0 results connect the decorrelation gap to measurable information cost. The 8.00 dB PSNR spread demonstrates that chroma subsampling is not uniformly safe across photographic content — a conclusion that has been assumed but not quantified at the per-image level. The degree of safety depends on the blue channel independence of the individual image, a metric now available for the full benchmark.

5 Conclusion

The BT.601 fixed-coefficient YCbCr transform achieves between 31.5% and 82.3% of the KLT optimum across the 24 images of the Kodak Lossless True Color Image Suite, with a mean efficiency of 55.4%. The dominant residual is the Cb–Cr chroma pair, which retains a mean absolute correlation of 0.616 — a structured, persistent redundancy that standard channel-independent encoder stages largely pass through. Standard 4:2:0 chroma subsampling imposes an information cost varying by 8.00 dB across the suite, correlating with per-image blue channel independence as established in Baetzel (2026a). The complete per-image measurements are provided as machine-readable JSON files for independent verification and further analysis.

References

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Source

Benchmark images: Kodak Lossless True Color Image Suite (r0k.us/graphics/kodak/). 24 images, 768×512 / 512×768, 24-bit RGB, lossless PNG. Released by Eastman Kodak Company for unrestricted usage.

Baseline characterization: Baetzel, J. (2026a). Per-Image Principal Component Decomposition of PCD0992.

Analysis: Python 3, NumPy, Pillow. All metrics computed on unprocessed pixel arrays in a single pass.